Fractals, Vol. 1, No. 4 (1993) 745–752

Physics

LARGE SCALE STRUCTURE OF INTERFACES: AN INVERSE METHOD

CHI-HANG LAM
Applied Physics Department, Yale University,
Box 2155 Yale Station, New Haven, CT 06520, USA

LEONARD M. SANDER
H.M. Randall Laboratory of Physics, The University of Michigan,
Ann Arbor, MI 48109-1120, USA

Abstract

We propose an inverse method to extract effective couplings and the renormalization group flow for growing interfaces. We apply it to discrete surface growth models in the Kardar-Parisi-Zhang universality class in $1+1$ dimensions and obtain the first measurement of a universal coupling constant. We consider interfaces not in the steady state. It may also be applicable to analyze experimental data and for other forms of interface growth.

1. INTRODUCTION

The growth of disorderly interfaces far from equilibrium has been intensely studied in recent years, and a number of remarkable advances have been made.[1,2] Examples include the Edward-Wilkinson (EW) equation describing sedimentation,[3] the Kardar-Parisi-Zhang (KPZ) equation describing vapor deposition and directed polymers,[4] and the diffusion-limited aggregation model.[5] In all of these cases the interface is specified by a set of kinetic rules. It is driven by noise to a steady state which has scaling features, and these features are seldom obvious from the microscopics of the rules. The most spectacular example of this is DLA where the large-scale structure appears to be at least approximately fractal, and no theory is known to predict this fact from the simple model.

A particularly interesting example is the Kuramoto-Svinshinsky equation of flame-front theory.[6] In this case, the interface is in fact specified by a deterministic equation which has an instability leading to cellular structures on the interface. It has recently been shown that on large scales, the correlations of the interface act exactly as if they were governed by the noise-driven KPZ equation. Again, there is scaling structure on these scales which is not in the least obvious from short-range considerations.

We have proposed[8] an inverse method to find essentially complete information about interface growth processes and to explicitly reveal the scaling. It closely resembles the spirit of Ma's Monte Carlo renormalization group, which computes numerically the effective block spin Hamiltonian from equilibrium spin configurations.[9] Our method can be considered as the dynamical and continuum counterpart and is applicable to both equilibrium and nonequilibrium systems. We can calculate directly the evolution equation for a given surface from simulation or possibly experimental data. The renormalization group flow of the equation can be obtained with *all* its parameters computed. We will review the method in this paper and give a new application to interfaces not in the steady state.

The method is as follows[8]: A time sequence of snapshots of a surface captures its dynamical properties up to the corresponding resolution. It is usually not difficult to guess a general candidate equation for the evolution containing all the possible terms consistent with symmetry. Our task is to discriminate among them. Given a snapshot of the surface, realizations of the equation with different coefficients give different predictions about the next snapshot. The correct equation is the one which, on the average, gives the best predictions over a large number of trials. It can thus be obtained by minimizing the errors of the predictions. Any terms with zero coefficients are proved to be absent.

We illustrated the method by working through applications to growth models in the KPZ universality class in 1+1 dimensions. The height-height correlation function of these surfaces in the steady state follows the scaling form: $\langle |h(x,t)-h(0,t')|^2\rangle \sim |x|^{2\alpha} f[|t-t'|/|x|^z]$. The roughness exponent α and the dynamical exponent z characterize the universality class and are thought to be the same for many microscopic models. Though these examples have been extensively studied previously, this novel technique is powerful enough to yield entirely new and important results as well as reproduce the known ones. We briefly review the method here.

The evolution of the surfaces we studied is known to be given by the KPZ equation[4]:

$$\frac{\partial h}{\partial t} = c + \nu \nabla^2 h + \frac{\lambda}{2}(\nabla h)^2 + \eta(x,t) \tag{1}$$

where the noise η has a Gaussian distribution and mean 0 and a correlator given by:

$$\langle \eta(x,t)\eta(x',t')\rangle = 2D\delta(x-x')\delta(t-t') \tag{2}$$

In addition, there is a lower wavelength cutoff, or spatial resolution, l. From dynamical renormalization group (DRG) calculations, for a given surface, ν and D depend on the resolution l of observation and, asymptotically for large l, we have[10,4]:

$$\nu(l) \sim l^{\alpha} \quad D(l) \sim l^{\alpha} \tag{3}$$

The parameters λ and $D(l)/\nu(l)$ are independent of l due to symmetry[11] and a fluctuation dissipation theorem[10] respectively. Both of them have been measured numerically[12–14] and, for some particular models, can be obtained analytically.[13,15]

We take the candidate evolution equation to be in a generalized form:

$$\frac{\partial h}{\partial t} = c + \nu\partial^2 h + \frac{\lambda}{2}(\partial h)^2 + a_4\partial^4 h + a_5\partial^2(\partial h)^2 + a_6\partial h\partial^3 h + a_7\partial^6 h + \cdots + \eta \qquad (4)$$

where all the differentiations on the right are with respect to x. Discretizing time, we have:

$$\frac{\Delta h(x,\, t)}{\Delta t} \simeq \mathbf{a}\cdot\mathbf{H}(x,\, t) + \eta(x,\, t) \qquad (5)$$

where $\mathbf{a} = (c,\, \nu,\, \frac{\lambda}{2},\, a_4, \ldots)$ and $\mathbf{H}(x,\, t) = (1,\, \partial^2 h,\, (\partial h)^2,\, \partial^4 h, \ldots)$ are vectors. We find the parameter vector $\mathbf{a}$ so that the equation best describes the snapshots. To measure $\mathbf{H}(x,\, t)$ numerically, we first simulate a given discrete model to obtain an initial discrete surface $h_d(x,\, t)$. We coarse-grain it by truncating the Fourier components of wavelength smaller than l. From the coarse-grained surface, we get $\mathbf{H}(x,\, t)$ which thus depends on l. The differentiations and multiplications involved are done in the Fourier and real spaces respectively. An additional coarse-graining is then carried out to maintain the cutoff which is shifted by the nonlinearities. We obtain m realizations of $h_d(x,\, t+\Delta t)$ by further growing the *same* surface $h_d(x,\, t)$ m times. Averaging gives the mean discrete profile at time $t+\Delta t$ from which we obtain the coarse-grained mean surface advance $\langle\Delta h(x,\, t)\rangle$, which depends on both the spatial and temporal resolutions l and Δt. For large m, Eq. (5) averages to:

$$\frac{\langle\Delta h(x,\, t)\rangle}{\Delta t} \simeq \mathbf{a}\cdot\mathbf{H}(x,\, t) \qquad (6)$$

The fitting can be done most conveniently with discrete data set. We look at N discrete points $(x_i,\, t)$ each separated spatially by l and denote $\mathbf{H}_i = \mathbf{H}(x_i,\, t)$ and $\langle\Delta h_i\rangle = \langle\Delta h(x_i,\, t)\rangle$. To enhance the statistics, the data set can be taken from M realizations of the pair $H(x,\, t)$ and $\langle\Delta h(x,\, t)\rangle$ generated from the corresponding M realizations of $h_d(x,\, t)$. Using a least squares fitting procedure, the mean square deviation of Eq. (6) from equality is:

$$\mathcal{D} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{\langle\Delta h_i\rangle}{\Delta t} - \mathbf{a}\cdot\mathbf{H}_i\right]^2 \qquad (7)$$

Minimizing $\mathcal{D}$ with respect to $\mathbf{a}$ gives the main equation of our inverse method, from which $\mathbf{a}$ can be found:

$$\mathbf{A}\mathbf{a} = \mathbf{b} \qquad (8)$$

where the matrix $\mathbf{A}$ and the vector $\mathbf{b}$ are defined by:

$$\mathbf{A} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{H}_i\otimes\mathbf{H}_i \quad \mathbf{b} = \frac{1}{N}\sum_{i=1}^{N}\frac{\langle\Delta h_i\rangle}{\Delta t}\mathbf{H}_i \qquad (9)$$

The symbol $\otimes$ denotes the vector outer product. Straightforward algebra gives the noise correlator as:

$$D = \frac{\Delta t l \mathcal{D}'}{2} \qquad (10)$$

where $\mathcal{D}'$ is defined like $\mathcal{D}$ in Eq. (7) with $\langle\Delta h_i\rangle$ replaced by an unaveraged Δh_i. We have assumed the δ-correlations in Eq. (2). We have now obtained expressions for all the parameters in the continuum formulation.

We applied the method to discrete models in the KPZ universality class. Two of them are single step models[17] in which the nearest-neighbor height differences of the surface is constrained to be ± 1. A particle of height 2 can be added to the aggregate only if the resulting step heights obey the constraint. The first model (SS I) is the original single step model, where there is one growth attempt at a random site per site per unit time.[17] The other one (SS II) adopts sublattice updating. During each time step, growth is attempted independently with 50% probability at all odd sites and then at all even sites.[15] We also simulated 2 variants of the ballistic deposition model with sublattice updating (BD I and BD II) with growth rules given by $h(x, t+1) = \max\{h(x, t)+\eta(x, t), h(x-1, t), h(x+1, t)\}$ and $h(x, t+1) = \frac{1}{2}[max\{h(x, t), h(x+1, t), h(x-1, t)\} + h(x, t)] + \eta(x, t)$[16] respectively, where the $\eta's$ are independent uniform random deviates.

For details of the fitting procedure see Ref. 8. The resulting measured model parameters for the SS I model are shown in Fig. 1. The statistical errors are small except noticeably

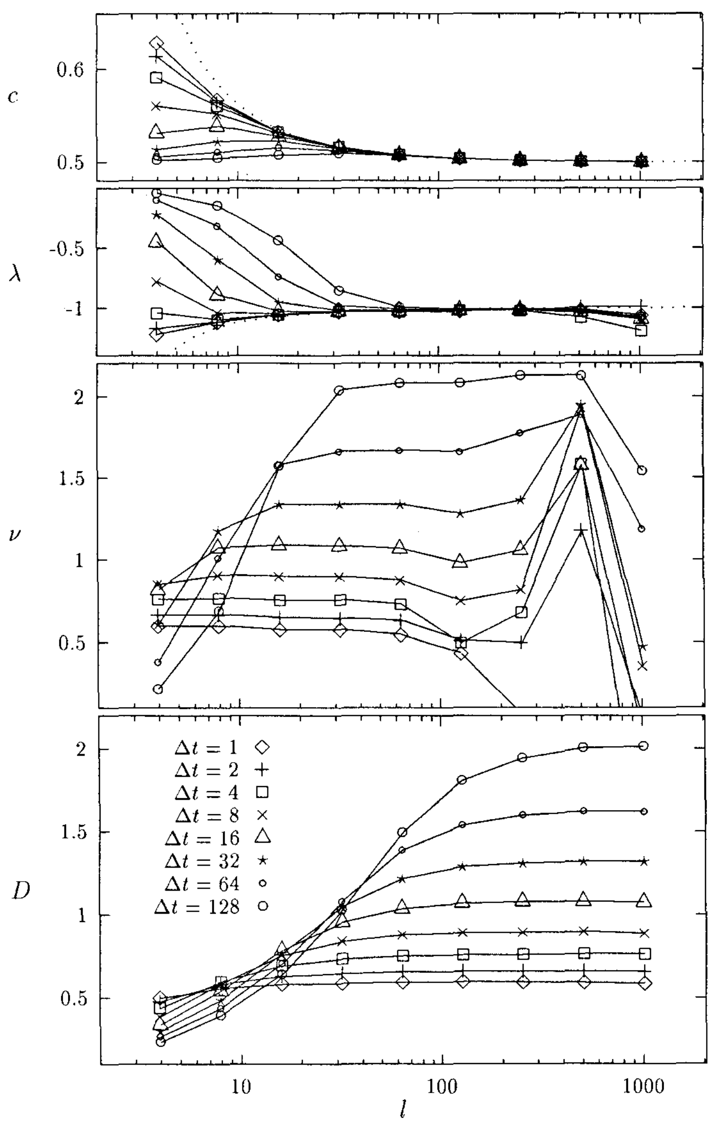

Fig. 1 Measured continuum parameters c, λ, ν and D as functions of the resolutions l and Δt for the SS I model.

Table 1 Numerical Values of λ, ν and D for $\Delta t = 128$, which give D/ν and $\bar{\lambda}$. Bracketed values are exact results

Model	λ	$\nu/\Delta t^{1/3}$	$D/\Delta t^{1/3}$	D/ν	$\bar{\lambda}^2$
SS I	−1.00 (−1.00)	0.413	0.399	0.966 (1.000)	5.66
SS II	−0.712 (−0.707)	0.254	0.236	0.929 (0.971)	5.68
BD I	1.86	0.253	0.0325	0.128	5.30
BD II	1.64	0.265	0.048	0.181	5.45

for ν at large l where the magnitude of this term is small compared with the others. For $l^z \lesssim \Delta t$, where $z = \frac{3}{2}$ in 1 + 1 dimensions, all the parameters are affected by the finite time difference approximation. From Fig. 1, both c and λ converge to the expected exact values with the "finite size corrections" $\frac{1}{2}[1-(l-1)^{-1}]^{-1}$ and $-[1-(l-1)^{-1}]^{-1}$ respectively (dotted lines).[18] The parameters ν and D admit renormalization and are much more interesting. Both ν and D converge as l increases and become functions of Δt only. Table 1 lists the numerical values of ν, D and D/ν at $\Delta t = 128$ for large l together with λ. Bracketed values are exact results.[13,15] The errors are estimated to be around 2 to 5%. The values of D/ν measured agree reasonably with the exact results but are systematically smaller because D, which is related to the error in the fitting, is underestimated due to the finite sample size. The agreement supports the δ correlation of the renormalized noise assumed in deriving Eq. (10).

Since the DRG calculations look at the long time behavior,[10] while we have finite observation time Δt, the asymptotic form in Eq. (3) is generalized to:

$$\nu(l, \Delta t) = l^\alpha f_1\left(\frac{\Delta t}{l^z}\right) = \Delta t^\beta f_2\left(\frac{\Delta t}{l^z}\right) \tag{11}$$

and similarly for D, where the early time exponent $\beta = \frac{\alpha}{z} = \frac{1}{3}$. The convergence of ν and D, which manifests itself as $\lim_{y\to 0} f_2(y) \to$ constant, corresponds to the fact that Δt controls the extent of the renormalization for sufficiently large l. This will be explained further later. With increasing Δt, the asymptotic scaling form [Eq. (11)] is expected and is demonstrated through the reasonable data collapse for ν at $\Delta t \gtrsim 32$ for the SS I model in Fig. 2. The result for D is similar. The success of the data collapse demonstrates that our method yields the correct exponents. By examining curves similar to Fig. 2 for different exponents, we can show that we must have $\beta = 0.33 \pm 0.01$ and $z = 1.50 \pm 0.04$.

For models in this class, asymptotic scaling is approached when a dimensionless coupling constant $\bar{\lambda}^2(\Lambda) = \lambda^2 D/\nu^3\Lambda$ flows to a finite universal fixed point upon renormalization in 1 + 1 dimensions, where $\Lambda = \pi/l$ is the upper momentum cutoff. Now the extent of renormalization is controlled by Δt and Λ should be replaced by an effective cutoff $c(\lambda^2 D/\nu)^{-1/3}\Delta t^{-1/z}$, where c is a universal constant and the remaining proportionality constants are from dimensional analysis. This leads to a redefined coupling constant $\bar{\lambda}^2(\Delta t) = [(\lambda^4 D^2/\nu^5)\Delta t]^{2/3}$. The numerical values are listed in Table 1. We obtained $\bar{\lambda}^2(\infty) = 5.5 \pm 0.2$, which is the first numerical estimate of the universal fixed point.

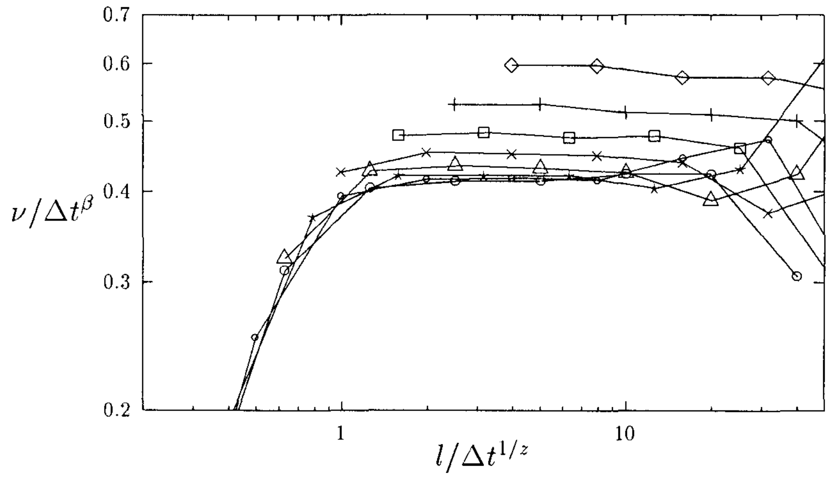

Fig. 2 Data collapse of the values of ν from Fig. 1.

Up until the present, we have been focusing on the evolution of surfaces in the steady state. This is also the regime to which renormalization group techniques were directly applied.[10] The early development of initially flat surfaces is less well understood. Most exact results become invalid. For example, the fluctuation dissipation theorem, which we needed[8] to prove that renormalized parameters should converge for large l, does not hold[10] and the probability functional of the surface is unknown analytically.

We applied the inverse method to unsaturated surfaces generated with the single step model (SS I). The simulation is similar to the previous case. However, instead of fully developed surfaces, $h_d(x, t)$ were prepared by depositing $t = 200$ layers of particles onto an initially flat substrate, which is, more precisely, zig-zag substrates following the ± 1 step height constraint. Their roughness is saturated only locally for segments shorter than the correlation length $\xi \simeq t^{1/z} \simeq 34$, where $z = \frac{3}{2}$. Figure 3 shows the computed parameters. For spatial resolution $l \ll \xi$, the parameters are the same as those in Fig. 1 for saturated surfaces. For $l \gtrsim \xi$, the renormalized coefficients are different in general. This is because, in the renormalization group sense, the short wavelength modes being integrated out do not follow the steady state distribution. They perturb the evolution of the remaining modes differently and contribute different renormalizations.

Since the fluctuation dissipation theorem does not hold, our parameters do not have to converge for large l. As shown in Fig. 3, ν turns negative for large l. The $\nu\nabla^2 h$ term then corresponds to an instability which can be understood as follows. It is known that the nonlinear coupling of the λ term speeds up the roughening of flat surfaces. This is because the r.m.s. surface width W scales with time like $W \sim t^{\beta}$, where the early time exponent $\beta = 1/3$ for the KPZ case is bigger than $\beta = 1/4$ for the Edward-Wilkinson case. After the short wavelength modes are integrated out, their contribution to the roughening of the remaining modes due to the nonlinear coupling have to be represented in the renormalized parameters. This amounts to an instability of the long wavelength modes. The same conclusion can also be obtained from a microscopic point of view. Note that a locally smoother (rougher) segments of the surface grow faster (slower), where we assumed negative λ but the argument is similar for positive λ. Hence the spatial variations in the local roughness are able to induce long wavelength fluctuations. Moreover, a particular pattern

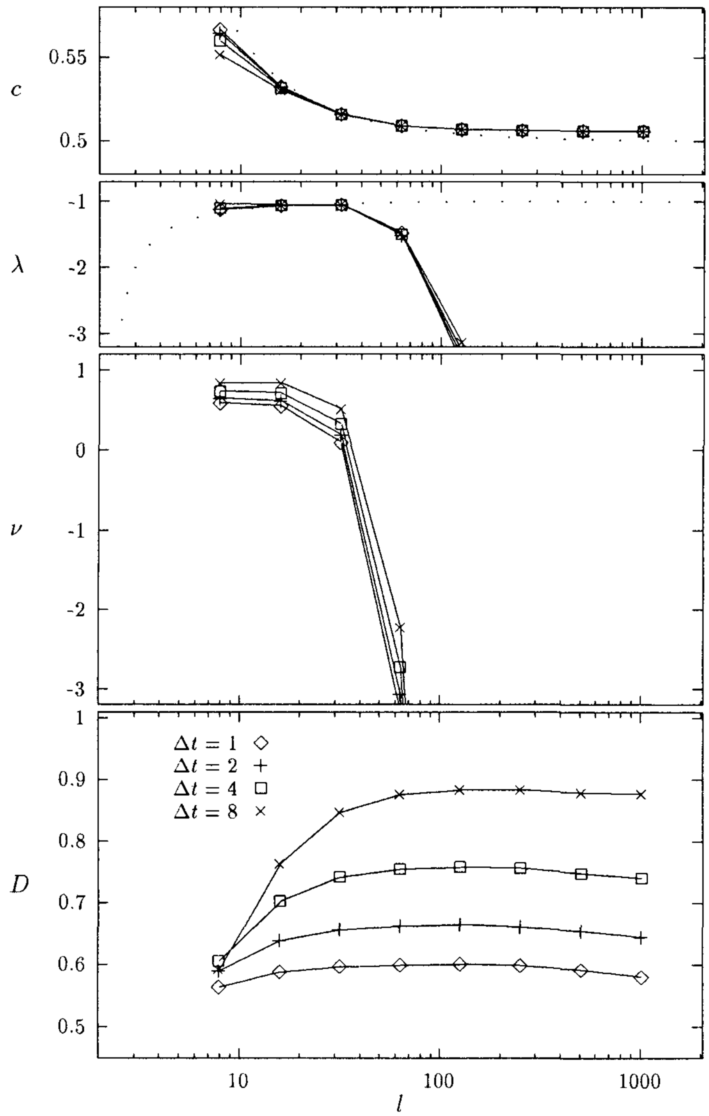

Fig. 3 **Measured continuum parameters c, λ, ν and D as functions of the resolutions l and Δt for the SS I model in the transient regime.**

of roughening persists for the period of time when the variations in the local roughness remain practically unchanged. This persistence allows further advance of already advanced local segments and is precisely an instability.

For the previous steady state case, λ is not renormalized due to a symmetry of the KPZ equation with respect to an infinitesimal tilt.[11] However, since we now consider the early development of a flat surface, the substrate orientation breaks the symmetry. This permits renormalization of λ as is evident in Fig. 3. The averaged growth speed c in Fig. 3 does not converge to the exact value $\frac{1}{2}$ for steady state surfaces. This is because a flatter surface is, on average, smoother and grows faster for the single step model with negative λ. The noise intensity D is indistinguishable in the two cases within our numerical accuracy.

We have demonstrated that when the spatial resolution l is comparable to the correlation length ξ, very different values of the parameters are obtained. In many cases, $l \ll \xi$ is desired since it is when the standard KPZ description holds. Then the preparation of fully roughened surfaces ($\xi = \infty$) is important. In our investigation[8] of $1+1$ dimensional surfaces in the KPZ class, they were generated directly according to the known distribution. In general, the distribution is unknown and deriving efficient algorithms to approximate the steady state surfaces is an important task.

We should note that the effective instability which we tracked in this simple case is rather like that of the Kuramoto-Sivashinsky case where a negative ν is eventually converted to a positive one after the surface develops enough. We think that an application to this case would be of particular interest.

Another case which we intend to study is the DLA model which has been a deep puzzle for a decade. In this case, even the formulation of a mean-field theory has proven to be a major challenge.[19] We can imagine trying a continuum formulation in that case as well, and trying to fit to data from extensive simulations which are now easily done.

In conclusion, our inverse method shows great promise for analysis of interface growth. Previous applications yielded the first computation of *all* the parameters in the KPZ equation at large scales. This is the most direct numerical verification of the applicability of the equation. The finiteness of the universal coupling constant has been proved numerically and its value computed. We have shown here that the interface not in the steady state acts as if it is unstable. The method should have very wide applicability.

ACKNOWLEDGMENTS

We thank D. S. Graff, F. Guinea, D. A. Kessler, D. Ko, O. Pla and R. B. Stinchcombe for interesting discussions. This work is supported by NSF Grant DMR91-17249.

REFERENCES

1. F. Family and T. Vicsek, *Dynamics of Fractal Surfaces* (World Scientific Pub. Co., 1991).
2. L. M. Sander, in *Solids Far From Equilibrium*, ed. C. Godreche (Cambridge, 1992).
3. S. F. Edwards and D. R. Wilkinson, *Proc. R. Soc. Lond.* **A381**, 17 (1982).
4. M. Kardar, G. Parisi and Y.-C. Zhang, *Phys. Rev. Lett.* **56**, 889 (1986).
5. T. A. Witten and L. M. Sander, *Phys. Rev. Lett.* **47**, 1400 (1981); *Phys. Rev.* **B27**, 2586 (1983).
6. Y. Kuramoto and T. Tsuzuki, *Prog. Theo. Phys.* **55**, 356 (1976); G. I. Sivashinsky, *Acta Astronautica* **6**, 569 (1979).
7. V. Yakhot, *Phys. Rev.* **A24**, 642 (1981), S. Zaleski, *Physica* **D34**, 427 (1989), C. Jayaprakash, F. Hayot and R. Pandit, *Phys. Rev. Lett.* **71**, 561 (1993).
8. C.-H. Lam and Leonard M. Sander, *Phys. Rev. Lett.* **71**, 561 (1993).
9. S. Ma, *Phys. Rev. Lett.* **37**, 461 (1976).
10. D. Forster, D. R. Nelson and M. J. Stephen, *Phys. Rev.* **A16**, 732 (1977).
11. E. Medina, T. Hwa, M. Kardar and Y.-C. Zhang, *Phys. Rev.* **A39**, 3053 (1989).
12. D. E. Wolf and L.-H. Tang, *Phys. Rev. Lett.* **65**, 1591 (1990).
13. J. Krug, P. Meakin and T. Halpin-Healy, *Phys. Rev.* **A45**, 638 (1992).
14. J. G. Amar and F. Family, *Phys. Rev.* **A45**, 5378 (1992).
15. L.-H. Tang, N. M. Forrest and D. E. Wolf, *Phys. Rev.* **A45**, 7162 (1992).
16. C.-H. Lam and L. M. Sander, *Phys. Rev. Lett.* **69**, 3338 (1992).
17. P. Meakin, P. Ramanlal, L. M. Sander and R. C. Ball, *Phys. Rev.* **A34**, 5091 (1986).
18. J. Krug and P. Meakin, *J. Phys.* **A23**, 987 (1990).
19. T. A. Witten and L. M. Sander, *Phys. Rev.* **B27**, 2586 (1983); R. Ball, M. Nauenberg and T. Witten, *Phys. Rev.* **A29**, 2017 (1984); E. Brener, H. Levine and Y. Tu, *Phys. Rev. Lett.* **66**, 1978 (1991).